PHISHING EMAIL DETECTION & AWARENESS SYSTEM

1. Aim

To analyze multiple phishing email samples, identify suspicious indicators, classify risks, and create awareness guidelines to prevent cyber attacks.

2. Introduction

Phishing is a type of cyber attack where attackers impersonate trusted entities to steal sensitive information like passwords, OTPs, and banking details.
These attacks rely on human emotions such as fear, urgency, and greed, making them highly effective.

This project analyzes 10 phishing email samples and explains how to detect and prevent such attacks.

3. Email Samples Used

The following types of phishing emails were analyzed:

- Account security warning (Email 1)
- Reward/lottery scam (Email 2 & 10)
- Banking fraud (Email 3 & 4)
- Internship/job scam (Email 5)
- E-commerce fraud (Email 6)
- Academic portal scam (Email 7)
- OTP phishing (Email 8)
- Google account compromise (Email 9)

4. Detailed Email Analysis

Example 1: Account Suspension Email (Email 1)

Subject: Urgent: Your Account Will Be Locked

Indicators:

- Generic greeting (“Dear User”)
- Urgent tone (creates panic)
- Suspicious URL (secure-account-verify[.]com)

- No official company details

Conclusion:

This email is a Phishing Attack designed to steal login credentials.

Example 2: Lottery Scam (Email 2 & 10)

Indicators:

- Unrealistic rewards (₹10,000 / ₹5,00,000)
- No verification or proof
- Suspicious domains (.xyz, .claim)
- Urgency to claim prize

Conclusion:

These are Scam/Phishing Emails targeting greed.

Example 3: Banking Phishing (Email 3 & 4)

Indicators:

- Fake bank/security message
- Fear-based language (account suspension)
- Suspicious URLs
- No official domain

Conclusion:

These emails attempt financial fraud and credential theft.

Example 4: Internship Scam (Email 5)

Indicators:

- Too-good-to-be-true offer
- Urgency ("within 12 hours")
- Suspicious link
- No official company details

Conclusion:

This is a Spear Phishing Attack targeting students.

Example 5: E-commerce Fraud (Email 6)

Indicators:

- Fake Amazon branding
- Payment issue alert
- Suspicious link
- Threat of cancellation

Conclusion:

This email aims to steal payment details.

Example 6: College Portal Scam (Email 7)

Indicators:

- Urgent password reset
- Fake domain (edu-login.com)
- Generic message

Conclusion:

This is a credential harvesting attack.

Example 7: OTP Phishing (Email 8)

Indicators:

- Requests OTP entry
- Fake verification link
- Urgent message

Conclusion:

This attack aims to bypass account security.

Example 8: Google Account Alert (Email 9)

Indicators:

- Fake security alert
- Suspicious recovery link

- Panic-inducing message

Conclusion:

This is a credential theft phishing attack.

5. Link & Domain Analysis

All email links are suspicious because:

- They do not match official company domains
- Use misleading keywords like “secure”, “verify”, “login”
- Use unusual extensions (.xyz, .net, .claim)
- Designed to mimic real websites

6. Email Header Analysis

Email headers help identify the real source of an email.

Findings:

- Sender email may be spoofed
- IP address may not match real organization
- Authentication checks (SPF/DKIM) may fail

Tools Used:

- Google Message Header Analyzer
- MXToolbox

7. Risk Classification

Email No.	Type	Risk Level
1	Account Phishing	High
2	Reward Scam	High
3	Banking Fraud	High
4	Banking Phishing	High
5	Internship Scam	High
6	E-commerce Fraud	High

Email No.	Type	Risk Level
7	Academic Scam	High
8	OTP Phishing	High
9	Credential Theft	High
10	Lottery Scam	High

8. Types of Phishing Attacks Identified

Type	Emails
Financial Phishing	3, 4
Reward/Lottery Scam	2, 10
Spear Phishing	5
E-commerce Scam	6
Credential Theft	7, 9
OTP Scam	8

9. How Phishing Attack Works

1. Attacker sends fake email
2. Creates urgency or excitement
3. User clicks malicious link
4. Fake website opens
5. User enters sensitive data
6. Attacker steals information

10. Prevention & Awareness Guidelines

Do’s

- Verify sender email carefully
- Hover over links before clicking
- Use official websites only

- Enable Two-Factor Authentication
- Report suspicious emails

Don’ts

- Do not click unknown links
- Do not share OTP/password
- Do not trust urgent messages blindly
- Do not download unknown attachments

11. Employee Awareness Tips

- Always verify suspicious emails
- Contact IT/security team when unsure
- Stay updated with cyber threats
- Use strong passwords

12. Conclusion

Phishing attacks are dangerous because they exploit human behavior rather than system weaknesses. By understanding phishing techniques and following safety practices, users can effectively prevent such attacks.

13. Tools Used

- Email Samples
- Google Header Analyzer
- MXToolbox
- Web Browser Tools